

ANNEX

DRAFT GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i. e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on

the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: **ECUADOR**
- Organization(s) or entity(s) responsible for the preparation of the report*:
Name: **Secretariat of Higher Education, Science, Technology and Innovation
SENESCYT**
Website: **<https://www.educacionsuperior.gob.ec/>**
Email address*:
Phone number*:
Please describe the role/mandate of your organization:

To exercise leadership over public policy on higher education, science, technology and ancestral knowledge and to manage its application, with a focus on the country's strategic development. To coordinate actions between the Executive and higher education institutions in order to strengthen academic, productive and social development. In the field of science, technology and ancestral

knowledge, to promote the training of advanced human talent and the development of research, innovation and technological transfer, through the development, execution and evaluation of policies, programs and projects.

- Officially designated contact person(s) who completed this survey:

Full name*: **Bianca Nathalie Palacios Pinargotti**

Position*: **Scientific Research Analyst**

Email address*:

- Full Name(s) of designated official(s) certifying the report:

Carlos Alberto Viña Castillo – Director of Scientific Research - Secretariat of Higher Education, Science, Technology and Innovation SENESCYT

- Brief description of the consultation process established for the preparation of the report:

The consultation process for the preparation of the report has been led by SENESCYT, the entity in charge of defining the actions, selecting the participants and establishing strategic collaborations in the areas of science, technology and innovation. Its role is fundamental to guarantee compliance with the commitments acquired within the framework of the UNESCO Recommendations.

As part of this process, the expert Francisco Silva Garcés, member of the Openlab Ecuador Foundation, a citizen laboratory that promotes dialogue and experimentation around digital culture, citizen participation and open knowledge, was consulted. In addition, he is part of the Global Steering Committee of Open Science, representing Latin America and the Caribbean.

In this context, SENESCYT and the Openlab Ecuador Foundation are the responsible entity and counterpart, respectively, for compliance with commitment C15 "Co-creation of the National Open Science and Strategic Research Policy", framed in the Second Open Government Action Plan (PAGA) led by the Open Government Directorate of the Presidency of Ecuador.

In order to prepare this report, the documents used in the development of the National Policy on Open Science and Strategic Research were analyzed, aligned with UNESCO recommendations and international standards that promote transparency, collaboration and equitable access to scientific knowledge.

The adoption of these principles will allow Ecuador to effectively integrate into the global scientific community, promoting international collaborations and strengthening the development of research in the country.

- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Openlab Ecuador Foundation

Website*: <https://openlab.ec/>

Email address*

Sector: ☐ Public ☐ Private ☒ Civil Society ☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO

If yes, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Participants, researchers, actors, institutions that participated in the working groups for the formulation of the National Policy on Open Science and Strategic Research:

For the working groups, delegates from the different actors of the National System of Science, Technology, Innovation and Ancestral Knowledge were requested. Below are the institutions that participated in the roundtables and formed part of the Open Science Core Group through their delegates:

- Secretariat of Higher Education, Science, Technology and Innovation SENESCYT
- Open Government Directorate of the Presidency
- Openlab Foundation
- National Intellectual Rights Service SENADI
- National Institute of Agricultural Research INIAP
- Ministry of the Environment, Water and Ecological Transition MAATE
- Ministry of Public Health MSP
- Council for the Assurance of the Quality of Higher Education CACES
- Central University of Ecuador UCE
- Higher Education Council CES
- Ecuadorian Corporation for the Development of Research and Academia CEDIA
- National Institute of Biodiversity INABIO
- Polytechnic School of the Litoral ESPOL
- National Institute of Meteorology and Hydrology INAMHI
- National Institute of Statistics and Census INEC
- National Polytechnic School EPN
- National Institute of Public Health Research INSPI

- **National Council for the Equality of Peoples and Nationalities**
- **Secretariat for the Management and Development of Peoples and Nationalities**
- **Universidad Andina Simón Bolívar UASB**
- **Latin American Faculty of Social Sciences FLACSO**
- **Ministry of Education**

The co-creation tables developed in three cities of the country had the participation of teachers, researchers, authorities from various universities, public and private research institutes, as well as delegates from public and private institutions. The participants are detailed below by city:

Quito

Higher Education Institutions

- **Simon Bolivar Andean University USAB**
- **Central University of Ecuador UCE**
- **International University of Ecuador UIDE**
- **International University SEK Ecuador**
- **National University of Loja UNL**
- **Indoamerica University UTI**
- **Equinoccial Technological University UTE**
- **National Polytechnic School EPN**
- **Polytechnic School of the Coast ESPOL**
- **University of the Armed Forces ESPE**
- **Pontifical Catholic University of Ecuador PUCE**
- **Tecnoecuadorian Higher Technological Institute**

Public Institutions

- **Higher Education Quality Assurance Council CACES**
- **Secretariat for the Management and Development of Peoples and Nationalities SGDPN**
- **General State Office FGE**
- **National Planning Secretariat SNP**
- **National Intellectual Property Service SENADI**
- **National Institute of Agricultural Research INIAP**

- **National Institute of Statistics and Census INEC**

- **Ministry of Public Health MSP**

Societies and Foundations

- **Latin American Society of Science and Technology SOLACyT**

- **INNOVACIENCIA Foundation**

- **INFOKRATOS Foundation (topics of research, dissemination and transfer of knowledge of conservation, sustainability, public policy and interculturality)**

- **National Union of Educators UNE**

- **Ecuadorian Center for Biotechnology and Environment**

Knowledge Networks

- **Knowledge and Academic Alliances Network Private Higher Institutes of Ecuador**

- **Ecuadorian Researchers Network REDI**

- **Latin American and Caribbean Scientific Researchers Network REDICALC**

- **SINERGIA Academic Network**

Guayaquil

Higher Education Institutions

- **Polytechnic School of the Litoral ESPOL**

- **University of Guayaquil**

- **Technical University of Babahoyo UTB**

- **Metropolitan University UMET**

- **Argos University Technological Institute**

- **Juan Bautista Aguirre Higher Technological Institute**

- **Simón Bolívar Higher Technological Institute ISTSB**

- **University of the Arts**

- **Catholic University of Santiago de Guayaquil UCSG**

- **Humane Technological University of Business**

- **Bolivarian University Higher Institute of Technology ITB**

- **Agrarian University of Ecuador UAE**

- **Technological Business University of Guayaquil UTEG**

Public Institutions

- **Public Institute of Aquaculture and Fisheries Research IPIAP**
- **Ministry of Public Health MSP**

Cuenca

Higher Education Institutions

- **University of Azuay**
- **University of Cuenca**
- **Catholic University of Cuenca**
- **Salesian Polytechnic University of Ecuador – Cuenca Campus**
- **National University of Education UNAE**
- **Technological University Higher Institute of Azuay ISUTA**

Public Institutions

- **National Institute of Public Health Research INSPI**

Companies and Foundations

- **Ecuadorian Corporation for the Development of Research and Academia CEDIA**

Knowledge Network

- **Hydrogen and Fuel Cell Network**

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

If yes, which language(s)? **Spanish**

- 1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

The approval and issuance of the National Policy on Open Science and Strategic Research is expected to take place in the second half of 2025, a key milestone for strengthening the scientific

ecosystem in the country. After its approval, a dissemination process will be carried out aimed at all actors in the National System of Science, Technology, Innovation and Ancestral Knowledge, with the aim of raising awareness about Open Science, highlighting its key elements, benefits and challenges. This process will allow greater commitment and participation of the scientific, academic and citizen community in the implementation of this policy, promoting transparency, collaboration and equitable access to knowledge.

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (*ref.: (i) [16.a](#), [16.b](#), [16.c](#), [16.d](#), [16.e](#), [16.h](#)*)

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

For the second half of 2025, the approval and issuance of the National Policy on Open Science and Strategic Research is expected.

Within the policy, the following guidelines are sought to be included, focused on incorporating the values and principles of open science in publicly funded research:

- Establish as a mandatory requirement for beneficiaries of national public funds for the development of academic-scientific research, that compliance with the FAIR and CARE principles, among others, for open research data be incorporated.

- Establish as a mandatory requirement for beneficiaries of national public funds for the development of scientific and academic research, that the resulting publications, their data and associated documents or resources be deposited in an open access digital repository, following the “green route” publication mechanism.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

YES / NO

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

Title: National Plan for the Social Economy of Knowledge, Creativity, Innovation and Ancestral Knowledge ESCCISA

Authority in charge of developing the policy: Secretariat of Higher Education, Science, Technology and Innovation SENESCYT

Entities involved in the development process:

Senescyt (Secretariat of Higher Education, Science, Technology and Innovation)

IES (Institutions of Higher Education)

IPI (Public Research Institutes)

INEC (National Institute of Statistics and Census)

SNP (National Secretariat of Planning)

MPCEIP (Ministry of Production, Foreign Trade, Investments and Fisheries)

MINTEL (Ministry of Telecommunications and the Information Society)

CACES (Council for the Assurance of the Quality of Higher Education)

CES (Council of Higher Education)

SENADI (National Service of Intellectual Property Rights)

MDT (Ministry of Labor)

MEF (Ministry of Economy and Finance)

ISTT (Technological Technical Higher Institutes)

MINEDUC (Ministry of Education)

CONGOPE (Consortium of Provincial Autonomous Governments of Ecuador)

HUB-iTT (Spaces for Innovation and Technology Transfer)

SP (Productive Sector)

STPN (Technical Secretariat of Peoples and Nationalities)

GADS (Decentralized Autonomous Governments)

MAG (Ministry of Agriculture and Livestock)

MAATE (Ministry of Environment, Water and Ecological Transition)

MERNNR (Ministry of Energy and Non-Renewable Natural Resources)

MSP (Ministry of Public Health)

ICBS (Incubators)

Year of adoption: 2023

Year of last update: --

Key areas/sections of the policy: Generate a knowledge-based development model, through the promotion and democratization of responsible and collaborative technological knowledge and innovative, which allows to respond to the social, environmental, productive and economic needs of the country, and is linked to global trends in S&T, taking into account the national context and from an inclusive approach.

Links to the website: <http://www.idearium.gob.ec/site/videos>

Anexo: 2.1.Plan_InnovaCIT

- does it include any of the following open science elements:
 - open access to scientific knowledge
 - open science infrastructure
 - open engagement of societal actors

o open dialogue with other knowledge systems

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) [16.a](#), [16.b](#), [16.c](#), [16.d](#), [16.e](#), [16.h](#)*)

YES / PARTLY / **NO**

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

SENESCYT is the entity responsible for fulfilling commitment C15: "Co-creation of the National Open Science and Strategic Research Policy", framed in the Second Open Government Action Plan (PAGA), led by the Open Government Directorate of the Presidency of Ecuador.

This commitment ended in December 2024 with 92% progress. Until its closure, the document proposed for public policy was still in the development phase and aligned with the requirements established by the National Planning Secretariat for the issuance of public policies in the country.

SENESCYT, as the entity responsible for the issuance of this policy, is carrying out a detailed review to ensure that its content does not generate overlaps or interfere with the competencies of the public sector institutions involved. Therefore, the guidelines of the National Planning Secretariat have been followed, with the aim of obtaining a public policy resolution of national scope.

The National Policy on Open Science and Strategic Research is projected to be approved and issued in the second half of 2025. This policy will be aligned with UNESCO recommendations and international standards, promoting transparency, collaboration and equitable access to scientific knowledge.

- 2.3 In your country, are there specific policy instruments^v that aim to promote open science?

YES/ UNDER DEVELOPMENT/ **NO**

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

SENESCYT is the entity responsible for fulfilling commitment C15: "Co-creation of the National Policy on Open Science and Strategic Research", framed in the Second Open Government Action Plan (PAGA), led by the Open Government Directorate of the Presidency of Ecuador.

This commitment ended in December 2024 with 92% progress. Until its closing, the document proposed for public policy was still in the development phase and aligned with the requirements established by the National Planning Secretariat for the issuance of public policies in the country.

The National Policy on Open Science and Strategic Research is projected to be approved and issued in the second half of 2025. This policy will be aligned with UNESCO recommendations and will promote the implementation of the values and principles of Open Science. In addition, it seeks to promote open science practices and support the cultural changes necessary for its consolidation. Various actors from the National System of Science, Technology, Innovation and Ancestral Knowledge, as well as representatives of civil society and the government, have participated in the policy formulation process. Their collaboration has been key to strengthening the social commitment to a more open, inclusive and transparent science.

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (*ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h*)

YES / **UNDER DEVELOPMENT** / NO

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

SENESCYT is in the process of formulating the National Policy on Open Science and Strategic Research, whose approval and issuance is projected for the second half of 2025.

Currently, the document is in the development phase and is aligned with the requirements established by the National Planning Secretariat for the issuance of public policies in the country. This process guarantees that the policy has a national scope, allowing its effective implementation in the science, technology and innovation system. In addition, it will be aligned with UNESCO recommendations and will promote the implementation of the values and principles of Open Science.

For the issuance of the policy, the participation of the following entities is necessary:

- **Council for the Assurance of the Quality of Higher Education (CACES).**- Public body that plans, coordinates and regulates the quality assurance system of higher education in Ecuador.
- **Higher Education Council (CES).**- One of the bodies that govern the system and whose mission is the planning, regulation and internal coordination of the Higher Education System of Ecuador, and the relationship between its different actors with the Executive Branch and Ecuadorian society.
- **Higher Education System Assembly (ASESEC).**- Body responsible for advising on the creation of an optimal higher education system.

These entities play fundamental roles in the regulation and improvement of higher education. In particular, the CES, CACES and SENESCYT are the bodies in charge of regulating this sector. Therefore, the public policy proposal will be sent to the highest authority of these

entities for review and validation of responsibilities, before the final issuance of the policy. This is essential, since the policy will affect all actors of the National System of Science, Technology, Innovation and Ancestral Knowledge, as well as the Higher Education System.

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

YES /UNDER DEVELOPMENT/ **NO**

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#); **Error! Marcador no definido.** included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

SENESCYT is in the process of formulating the National Policy on Open Science and Strategic Research, which will impact all actors in the National System of Science, Technology, Innovation and Ancestral Knowledge, as well as the Higher Education System. This includes universities, research institutes and other entities linked to scientific and academic research.

- 2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (ref.: [\(ii\) 17.j, \(iii\) 18.j, \(iv\) 19.c](#))

YES / UNDER DEVELOPMENT / **NO**

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

The approval and issuance of the National Policy on Open Science and Strategic Research is expected to take place in the second half of 2025. This policy will include guidelines and strategies aimed at developing financing mechanisms, based on the budgets assigned to the entity and, to the extent possible, seeking additional sources of resources.

- 2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

YES /**NO**

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The allocation of budgets by entities has not, to date, integrated open science practices into their strategies and guidelines for funding research. Therefore, the National Policy on Open Science and Strategic Research aims to include guidelines and strategies aimed at developing funding mechanisms for research that incorporate open science practices. These mechanisms will be based on the budgets allocated to the entities and, when possible, will seek additional sources of resources.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

YES / NO

If **yes**, please provide details and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In the area of Open Science, SENESCYT has been a founding member and political authority of LA Referencia (Federated Network of Institutional Repositories of Scientific Publications) since 2010. This network supports national open access strategies in Latin America, through a platform with interoperability standards, which allows sharing and giving visibility to the scientific production generated by higher education and scientific research institutions.

Since 2019, SENESCYT and the Ecuadorian Corporation for the Development of Research and Academia (CEDIA) have maintained a partnership through a technical-scientific cooperation agreement, which is renewed every two years, in order to continue supporting the operational management of LA Referencia. This collaboration has given rise to various national initiatives in the field of institutional repositories and open access, including the creation of a network of digital repositories, the metadata application project and training in repository standardization.

Links:

<https://cedia.edu.ec/>, <https://rraae.cedia.edu.ec/vufind/> ;
<https://www.lareferencia.info/es/nodos/ecuador> .

In addition, the National Policy on Open Science and Strategic Research will establish guidelines focused on establishing alliances and collaboration agreements with other countries and local or international organizations that promote open science in accordance with UNESCO recommendations.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

YES / UNDER DEVELOPMENT / **NO**

If **yes or under development**, please provide details and links as relevant.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

In the case of Ecuador, the latest official data were collected in the National Survey of Science and Technology Activities (ACTI) (INEC - Senescyt, 2016) corresponding to 2014 and recording an investment in R&D with respect to GDP of 0.44%, a percentage that is below the minimum investment established in the COESCCI (2016) which is 0.55% of GDP.

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

In the case of Ecuador, the latest official data was collected in the National Survey of Employment, Unemployment and Underemployment (ENEMDU) in 2024, which records that 66% of households in the country have access to the Internet.

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\) 18.c](#))

YES / NO

If **yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

With the aim of promoting the formation of collaborative networks of scientific, academic, cultural or social innovation research, which encourage the national and regional circulation and transfer of knowledge, the Secretariat of Higher Education, Science, Technology and Innovation SENESCYT offers the service of registering knowledge networks of the country of an academic, cultural, research and/or innovation nature.

The networks can be formed by the following actors:

- Decentralized autonomous governments within the scope of their competences;
- Higher education institutions, public and private, national and international ;
- Scientific research entities;
- International research centers;
- Academies of sciences;
- Natural persons, legal entities and other types of associations related to activities of the social economy of knowledge, creativity and innovation in all sectors of the economy, including the socio-productive sector and the popular and solidarity economy sector;
- Indigenous communities, peoples, and nationalities, through their contributions in the field of knowledge;
- Public institutions, public companies and other entities related to responsible research, strengthening of human talent, knowledge management, science, technology, social innovation, traditional knowledge and creativity, both at central and decentralized levels.

Information on registered networks can be accessed at the following link <https://www.educacionsuperior.gob.ec/sic-registro-de-redes/>

3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country?
(ref.: (iii) [18.d](#), [18.e](#), [18.k](#), [18.l](#), [18.m](#))

YES / UNDER DEVELOPMENT / NO

If **yes** or **under development**, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Repositories of Higher Education Institutions.

In Ecuador, the norm that requires universities to have a repository of theses was established in 2011, when the Higher Education Council (CES) approved the Technical Regulations for the management of degree projects. This regulation was part of an effort to regulate the academic degree process in the country's universities, promoting open access to research projects carried out by students.

In addition, the Organic Law of Higher Education (LOES), which was approved in 2010, also establishes principles related to open access to academic and scientific information, strengthening transparency and collaboration in the educational field.

Since then, universities in Ecuador are required to implement institutional repositories where theses, thesis works and other relevant academic documents are archived, ensuring that research results are publicly and digitally accessible, so they are non-commercial infrastructures and that in compliance with the 2021 Recommendation on Open Science, they comply with the FAIR and CARE principles.

The Mintel open data portal.

It is a platform for the exchange of knowledge from various areas available to citizens where research data is shared. It is a non-commercial platform that complies with the FAIR and CARE principles.

Link: <https://datosabiertos.gob.ec/>

SciELO-Ecuador Platform.

SciELO Ecuador is an open access platform that is part of the SciELO (Scientific Electronic Library Online) network, a system that promotes free and open access to scientific and academic production in Latin America, the Caribbean, Spain and Portugal. The platform

allows researchers, academics and students to access a wide range of scientific journals and academic articles from various areas of knowledge, such as social sciences, natural sciences, health, engineering, among others.

Through SciELO Ecuador, we seek to promote free access and the dissemination of scientific knowledge, support the quality and visibility of Ecuadorian scientific publications and improve international collaboration between researchers from different countries. It also plays a key role in the development of open science, by offering users the possibility of accessing, reading and sharing works without economic or geographical barriers, making it a non-commercial platform that, in compliance with the 2021 Recommendation on Open Science, complies with the FAIR and CARE principles.

Link: <http://scielo.senescyt.gob.ec/>

Open Access Repository Network of Ecuador RRAAE.

National initiative on the subject of Institutional Repositories and Open Access. It manages, decentralizes, organizes, and preserves open access digital content generated by the institutions of the academic-scientific community of the country. The RRAAE is the national node that is part of the Federal Network of Institutional Repositories of Scientific Publications of Latin America – La Referencia. It is a commercial platform but it complies with the FAIR and CARE principles.

Link: <https://rraae.cedia.edu.ec/vufind/>

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (*ref.: (iii) 18.e, 18.f*)

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science¹, stage of research, accessibility to all the relevant users in the region or internationally).

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES / NO

Ecuador hosts the following information technology infrastructures for open science:

- Ecuadorian Open Access Repository Network RRAAE

The RRAAE Network (Ecuadorian Academic and Research Repository Network), managed by CEDIA (Ecuadorian Corporation for the Development of Research and Academia), was created in 2011 in Spanish. Until 2025, it has 58 participating institutions. The RRAAE is designed to interconnect the institutional repositories of universities and research centers in Ecuador, facilitating access and dissemination of scientific and academic production generated in the country. Its purpose is to promote open access, knowledge exchange and visibility of research work carried out in Ecuador, which is aligned with the principles of open science, which seek to make research and knowledge freely accessible.

The RRAAE has a federated structure, meaning that it connects different repositories so that users can access a broad set of content from a single platform, regardless of which institution the research was generated at. This feature makes it similar to other federated infrastructures

at an international level, such as the repositories of the SciELO Network or OpenAIRE in Europe.

Although it is mainly focused on Ecuador, the RRAAE has the capacity to integrate with other repository and research networks at a regional and international level, allowing greater visibility and global access to Ecuadorian scientific production. This type of infrastructure facilitates access, exchange and collaboration in the field of scientific research at a national and international level.

Link: <https://rraae.cedia.edu.ec/vufind/>

- SciELO-Ecuador

Ecuador is part of the SciELO network since 2015. SciELO Ecuador is an open access portal that hosts the scientific publications of Ecuadorian journals. It is aligned with the principle of open science and its goal of making science accessible to all.

SciELO Ecuador is part of an international network that connects repositories from various countries. This federation of portals allows Ecuadorian publications to be accessible at a regional and international level, promoting the exchange of knowledge and scientific collaborations. In addition, it facilitates interoperability with other scientific platforms and databases, allowing users to access research and publications from different countries within the same network.

Link: <http://scielo.senescyt.gob.ec/>

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science', stage of research, accessibility to all the relevant users in the region or internationally).

Ecuador participates in the following North-South-South collaborations:

- SciELO-Ecuador

Ecuador is part of the SciELO network since 2015. SciELO Ecuador is an open access portal that hosts scientific publications from Ecuadorian journals. It is aligned with the principle of open science and its goal of making science accessible to all.

SciELO Ecuador is part of an international network that connects repositories from various countries. This federation of portals allows Ecuadorian publications to be accessible at a regional and international level, promoting the exchange of knowledge and scientific collaborations. In addition, it facilitates interoperability with other scientific platforms and databases, allowing users to access research and publications from different countries within the same network.

Link: <http://scielo.senescyt.gob.ec/>

- Latindex

Latindex (Regional Online Information System for Scientific Journals from Latin America, the Caribbean, Spain and Portugal) is a network that groups and classifies scientific journals from countries in Latin America, the Caribbean, Spain and Portugal, with the aim of promoting visibility and open access to scientific production in the region. It is a platform that optimizes the use of infrastructure and joint strategies for open scientific platforms shared at multinational, regional and national levels. Ecuador has been part of this System since 2001.

Link: <https://www.latindex.org/latindex/Solr/Busqueda?pais=Ecuador>

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (*ref.: (iv) 19.a, 19.c*)

YES / NO

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Ecuador, there are initiatives carried out by members of civil society, such as the Openlab Ecuador Foundation, the institution's counterpart for the creation of the National Policy on Open Science and Strategic Research.

This entity has developed the Discovering Open Science Conference, an event that was born in Ecuador in 2020 in the midst of the COVID19 pandemic, and became a regional event with support, collaboration and contributions from the Ibero-American Open Science Community.

Until 2024, four editions of this event have been held, which aims to spread open science to all possible corners and social actors. These conferences take place in several universities in Ecuador and are broadcast online for the entire region. They feature the participation of speakers, researchers and leaders of Open Science from Ibero-America.

In addition, in 2024, the LATAM Open Science Forum was held in the city of Quito - Ecuador during the month of September 2024. This event brought together academics, researchers, students and activists interested in the challenges of Open Science in the region and the continent. In addition, keynote presentations, talks, workshops and networking spaces with the open science community were held. The objective of the event was:

- Celebrate the achievements and recognize the most outstanding projects and initiatives in open science in Latin America.

- Promote collaboration and the exchange of ideas between the Latin American and international scientific community. Create a space conducive to the exchange of ideas, the generation of work networks and the establishment of new strategic alliances.

- Make visible the importance and highlight the crucial role of open science for the scientific, technological and social development of the region, in coherence with the Sustainable Development Goals (SDG).

- Promote the adoption of open science practices in universities and research institutions in the region.

- Discuss the challenges and opportunities that open science presents for the scientific and technological development of Latin America.

- Develop strategies to promote open science as a tool for inclusion, equity and social innovation.

Link to the Openlab Ecuador Foundation: <https://openlab.ec/>

Link to information on the event Jornadas Descubriendo la Ciencia Abierta: <https://openlab.ec/cienciaabierta2023#no-back>

Link to information on the Foro Latam 2024: <https://openlab.ec/taxonomy/term/80>

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / NO

If **yes**, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In 2023 and 2024, SENESCYT's strategic partners have carried out initiatives such as webinar training for policy makers on how to integrate the fundamental values and principles of open science into STI policies and strategies, particularly on Recommendations for the generation of Open Science public policy in Ecuador and the management of research data.

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

YES / **NO**

If **yes**, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

SENESCYT is in the process of formulating the National Policy on Open Science and Strategic Research, whose approval and issuance is projected for the second half of 2025. Within this policy, guidelines will be proposed focused on promoting that the country's Interinstitutional Quality Assurance System incorporate criteria in the evaluation of the quality of scientific research; of teaching and research staff; and of higher education institutions, which positively value the implementation of the principles, values, and practices of open science. These modifications in the Quality Assurance System require time for Higher Education institutions to implement them effectively. In addition, the evaluations of these institutions must be planned at least four years in advance. Therefore, the proposed guidelines would be applied in the next evaluation cycle, starting in 2028.

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES/ PARTLY/ **NO**

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

YES / NO

If **yes**, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

SENESCYT is in the process of formulating the National Policy on Open Science and Strategic Research, whose approval and issuance is projected for the second half of 2025. Within this policy, guidelines will be proposed focused on promoting and supporting scientific communication in accordance with the values and principles of open science.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20j](#))

YES / NO

If **yes**, please provide more information and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Within the framework of the formulation of the National Policy on Open Science and Strategic Research, whose approval and issuance are planned for the second half of 2025, various actors participated in the working groups, including the Council for Quality Assurance of Higher Education (CACES) and the Council of Higher Education (CES). The CACES, as a public body in charge of planning, coordinating and regulating the quality assurance system in higher education in Ecuador, and the CES, responsible for the planning, regulation and coordination of the Higher Education System, agreed to incorporate criteria in the evaluation of the quality of scientific research, of teaching and research staff, and of higher education institutions. These criteria will recognize and positively value the adoption of the principles, values and practices of open science.

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

YES / **NO**

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Currently, there are no specific clauses that recognize open science practices in the evaluation and career progression processes. This is because open science is a relatively new approach, the knowledge and application of which are still being socialized. In this context, the formulation of the National Policy on Open Science and Strategic Research, whose approval and issuance are planned for the second half of 2025, contemplates the inclusion of guidelines and strategies aimed at recognizing and valuing these practices in the academic and professional fields.

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h\)](#))

YES / **NO**

If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Currently, there are no initiatives that recognize or encourage open science practices. This is because open science is a relatively new approach, the knowledge and application of which are still being socialized. In this context, the formulation of the National Policy on Open Science and Strategic Research, whose approval and issuance are planned for the second half of 2025, contemplates the incorporation of guidelines and strategies aimed at valuing and promoting these practices both in the academic field and in the participation of social actors.

- 5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\) 20\)](#))

YES / **NO**

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21\)](#))

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

If **yes**, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Ecuador, there are various initiatives that recognize and value the contributions of social actors in citizen science projects and collective scientific collaboration. Below are some notable examples:

1. **Citizen Science for Biodiversity Conservation: The Wildlife Conservation Society (WCS) Ecuador**, in collaboration with the National Institute of Biodiversity (INABIO), has promoted citizen participation in biodiversity conservation. They have organized workshops and activities in which they train students and the general public in the use of applications such as iNaturalist and Ictio for the registration and identification of species. These tools allow citizens to actively contribute to the monitoring and conservation of biodiversity in their communities. fias.org.ec+2ecuador.wcs.org+2programs.wcs.org+2

2. **Darwin200 Project**: This global initiative seeks to inspire young leaders in environmental conservation. During its time in Ecuador, the project collaborated with scientists and students from the Universidad San Francisco de Quito (USFQ) and the Galapagos Science Center. Together, they conducted research and observations of species in the Galapagos Islands, encouraging the active participation of the academic and local community in scientific projects of international scope. biodiversity.gob.ec+1noticias.usfq.edu.ec+1biodiversidad.gob.ec+1

3. **Citizen Science Training in Educational Institutions**: The Foundation for Research and Social Development (FIAS) has conducted citizen science training aimed at educational institutions in Ecuador. These activities teach students and teachers to use platforms such as iNaturalist to record and share observations of local biodiversity, promoting a participatory scientific culture from an early age. <https://fias.org.ec/capacitacion-eds/>

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (*ref.: (vii) 22.a*)

Ecuador has made progress in promoting and facilitating international scientific collaborations on open science through various policies, initiatives and cooperation agreements. These efforts are aligned with the UNESCO Recommendation on Open Science (2021) and seek to guarantee open access to knowledge, strengthen scientific cooperation and encourage citizen participation in research.

- **Formulation of the National Policy on Open Science and Strategic Research:**

Ecuador is in the process of developing its National Policy on Open Science and Strategic Research, whose approval and issuance are planned for the second half of 2025. This policy will establish guidelines for international collaboration, open access to data and research, and the integration of global standards in open science.

- **Participation in international networks and programs:** Ecuador is part of regional and international organizations and consortia that promote collaboration in open science, such as the Federated Network of Institutional Repositories of Scientific Publications of Latin America (LA Referencia), which facilitates access to the country's scientific production; Ecuadorian universities and research centers have participated in Horizon Europe, the European Union program that funds research in open science and global cooperation; Cooperation agreements have been signed with international organizations, such as UNESCO and ECLAC, to strengthen open science and the transfer of knowledge.

- **Development of infrastructure for open access and scientific data:** Ecuador has institutional digital repositories in universities and research institutions, promoting open access to scientific publications and data; The Digital Library of Ecuador (<http://bibliotecasdeecuador.com/>) guarantees the visibility of scientific knowledge produced in the country and its free access internationally.

- **Promotion of Citizen Science and the participation of social actors:** Citizen science projects have been developed, such as the use of platforms such as iNaturalist and GBIF Ecuador for the collection of data on biodiversity; Educational programs and workshops have encouraged the participation of society in the generation and dissemination of scientific knowledge.

- **Incentives for scientific collaboration in open science:** Evaluation criteria have been included in calls for funding for research that favor open access publication and international collaboration; Universities have promoted the adoption of open licenses in scientific publications and have facilitated the interoperability of their repositories with international systems.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b](#), [22.e](#), [22.f](#))

YES / NO

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Ecuador has demonstrated a strong commitment to promoting open science and cross-border collaboration through various initiatives and strategies. Below are some of the most notable actions:

1. **Co-creation of the National Open Science and Strategic Research Policy:** Within the framework of the Second Open Government Action Plan (2022-2024), the Secretariat of Higher Education, Science, Technology and Innovation (SENESCYT) and the Openlab Ecuador Foundation committed to developing a national open science policy. This effort seeks to

integrate the voices of multiple actors, including academia, the public sector and civil society, to foster inclusive and participatory governance in open science.

2. International Partnerships for Strengthening Open Science: Ecuador has established collaborations with international networks such as AmeliCA, with the aim of improving scientific visibility and adopting open science practices. This alliance focuses on strengthening research capacity through the training and professionalization of researchers, as well as on the preparation of reports on scientific production in the country.

3. Participation in Regional Open Science Networks: The country is an active member of LA Referencia, a network that integrates open access repositories in Latin America. This participation facilitates cross-border collaboration and the exchange of best practices in open science, promoting interoperability and visibility of Ecuadorian scientific production at a regional level.

4. Organization of Events for the Dissemination of Open Science: Since 2020, Ecuador has hosted the "Jornadas Descubriendo la Ciencia Abierta", an annual event that seeks to disseminate the principles and practices of open science among various social actors. This space has had the collaboration of the Ibero-American open science community, strengthening international cooperation and the exchange of experiences.

5. Participation in International Forums: Ecuador has been present at events such as the LATAM Open Science Forum, co-organized by the Openlab Ecuador Foundation and the Ecuadorian Corporation for the Development of Research and Academia (CEDIA). These forums emphasize the relevance of open science as a driver of social change and promote new strategic alliances at the regional level.

These initiatives reflect Ecuador's commitment to promoting open science and cross-border collaboration, aligning with international recommendations to foster more inclusive and accessible science.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: [\(vii\) 22.c](#))

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Ecuador has shown an active commitment to promoting open science and has participated in regional and international debates on funding mechanisms for this area. Below are some of the most notable initiatives:

1. Participation in the LATAM Open Science Forum 2024: In September 2024, Ecuador hosted the LATAM Open Science Forum, an event that brought together academics, researchers, students, and activists to discuss the challenges and opportunities in open science in Latin America. This forum included discussions on the creation of regional funding mechanisms that facilitate collaboration and knowledge sharing between countries.

2. Participation in Regional Networks: The country is an active member of networks such as LA Referencia, which integrates open access repositories in Latin America. This participation facilitates cross-border collaboration and the exchange of best practices in open science, promoting the development of joint policies and strategies.

These actions reflect Ecuador's commitment to promoting open science and its participation in debates to establish funding mechanisms that encourage regional and international collaboration in this field.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/NO

If yes, please provide more information.

-
- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
- xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
- xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

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